Mateus Leite Pinho

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SUMMARY

Computer Science Master student and System Administrator at RNL, Técnico, managing Windows infrastructure. Experienced in NixOS for self-hosting. Developing full-stack web applications using Python and TypeScript with AI/LLM integration.

EXPERIENCE

System Administrator

Instituto Superior Técnico **June 2023 - Present, Lisbon, Portugal**

- Maintained 100+ lab computers, 10 servers, and 30 VMs alongside five grantees.
- Maintained over 10 Windows Server environments utilising Active Directory and Group Policy to support secure and reliable domain infrastructure.
- Engineered NixOS configuration, ensuring smooth functioning of critical infrastructure.
- Developed Grafana dashboards and integrated monitoring endpoints into Prometheus.

Teaching Assistant - Cyber-Security Forensics

Instituto Superior Técnico **August 2024 - January 2025, Lisbon, Portugal**

- Co-developed the practical part of the course with five other teaching assistants, designing a three part project focused on steganography, disk forensics, and network analysis.
- Led practical lab sessions to 20 students, providing mentoring and troubleshooting to students on forensics concepts.
- Designed forensic analysis modules requiring students to perform deep-dive disk and backup image investigations to identify system artifacts.

PROJECTS

NixOS Configuration

github.com/footvaalvica/nix-config • October 2024 - Present

- Maintained NixOS infrastructure-as-code with five modular host configs and reusable service definitions for consistent deployments.
- Ran five self-hosted services (Nextcloud, Immich, Home Assistant, Firefly III, Overleaf) with monitoring and health checks using Grafana and Prometheus.
- Built GitHub Actions CI/CD to auto-update Nix flake inputs and deploy NixOS systems via deploy-rs.

EDUCATION

Master In Computer Science and Engineering

Instituto Superior Técnico • Lisbon, Portugal • 2026 (Expected)

- Master Thesis: **"A new learning platform for Japanese learning, leveraging AI and gamification features"**. Architected a full-stack learning platform using FastAPI (Python) and React (TypeScript), integrating Google GenAI for real-time conversation.

Bachelors in Computer Science and Engineering

Instituto Superior Técnico • Lisbon, Portugal • 2023

SKILLS

- Programming Languages: Python, Typescript, Javascript, C++, Java, SQL
- Tools: Docker, Git, Figma, Nix, Ansible, Grafana, Prometheus, FastAPI, React, Celery,
- Systems: Linux System Administration, Windows System Administration
- User Interface (UI) Design
- Languages: Portuguese (Native), English (C2), Japanese (Beginner)

INVOLVEMENT

Sound Designer and Composer

Instituto Superior Técnico. • GameDev Técnico • February 2021 - February 2026

- Contributed to the long term project, The Ecosphere Project, producing the sound effects and composing the soundtrack.
- Led sound design and music composition for eight game jam projects, delivering music and sound effects under tight 48-to-72-hour deadlines.
- Composed and produced the original soundtrack for EDUCE, reaching a milestone of 5,000+ streams on Spotify.